



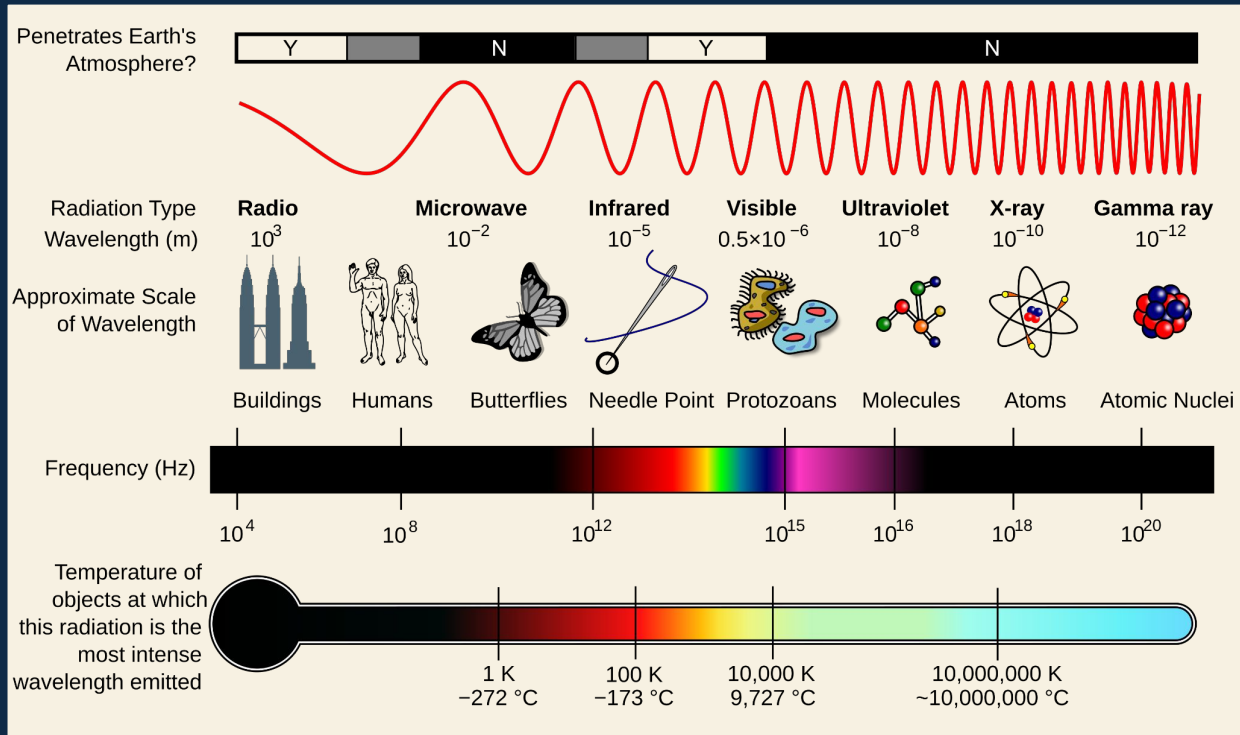
Intro to Gravitational Waves: Detection Across the Spectrum

Alexander W. Criswell

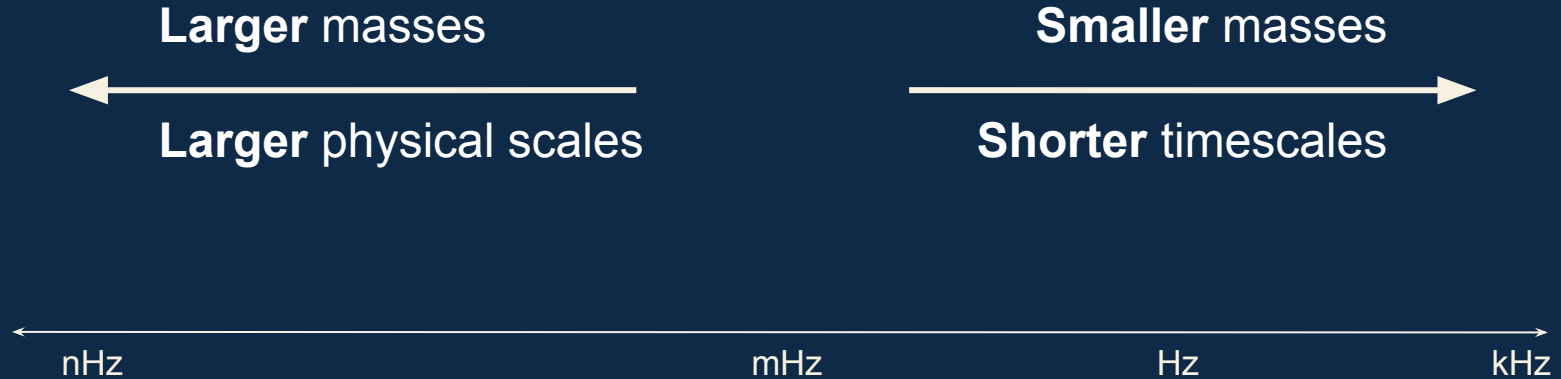
2026 VIPER Summer School | Vanderbilt University | 07.20.26

What to Expect

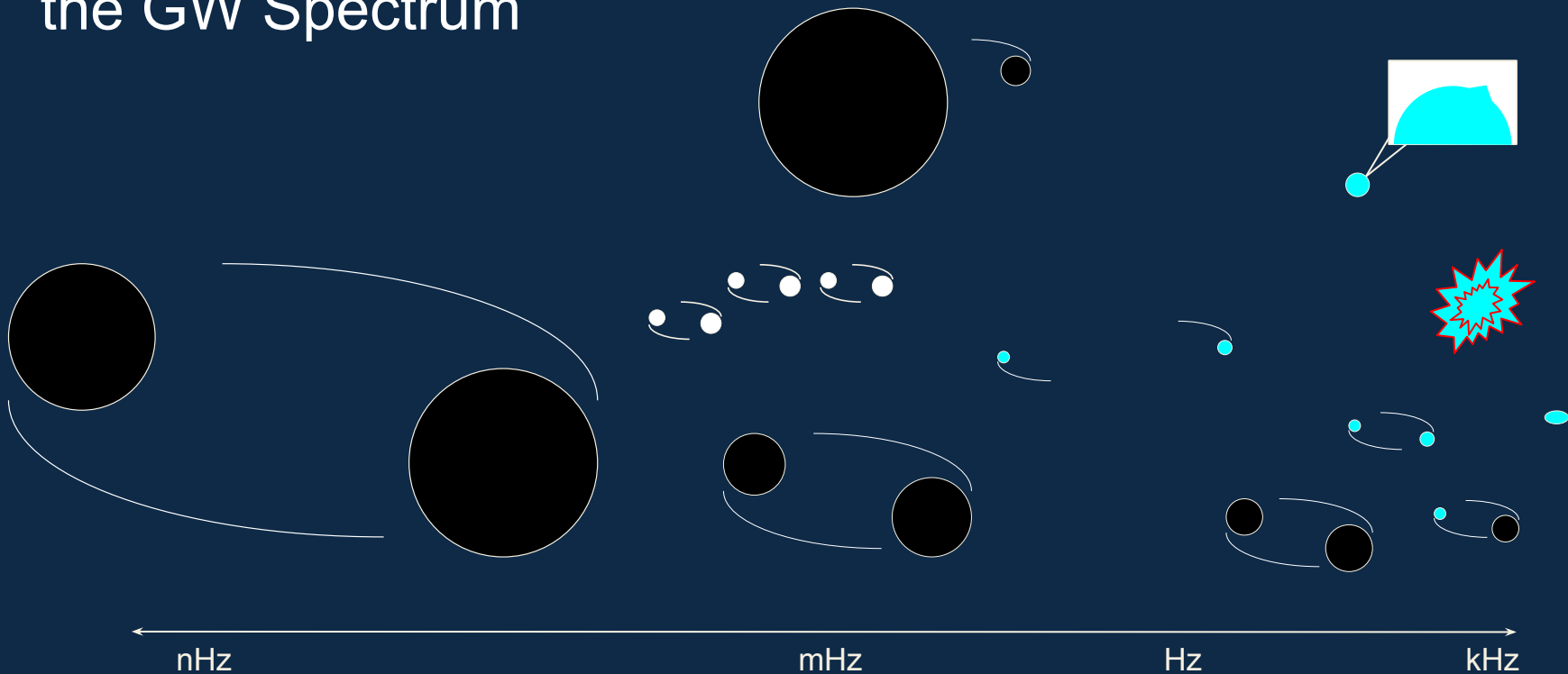
- A primer on the gravitational-wave (GW) spectrum
- Basic principles of GW detection
- A brief history of the field
- Current GW detection efforts & sources of interest
- Future concepts across the spectrum!



The GW Spectrum



Sources across the GW Spectrum



How do we detect GWs?

$$\text{strain} = \Delta L / L$$

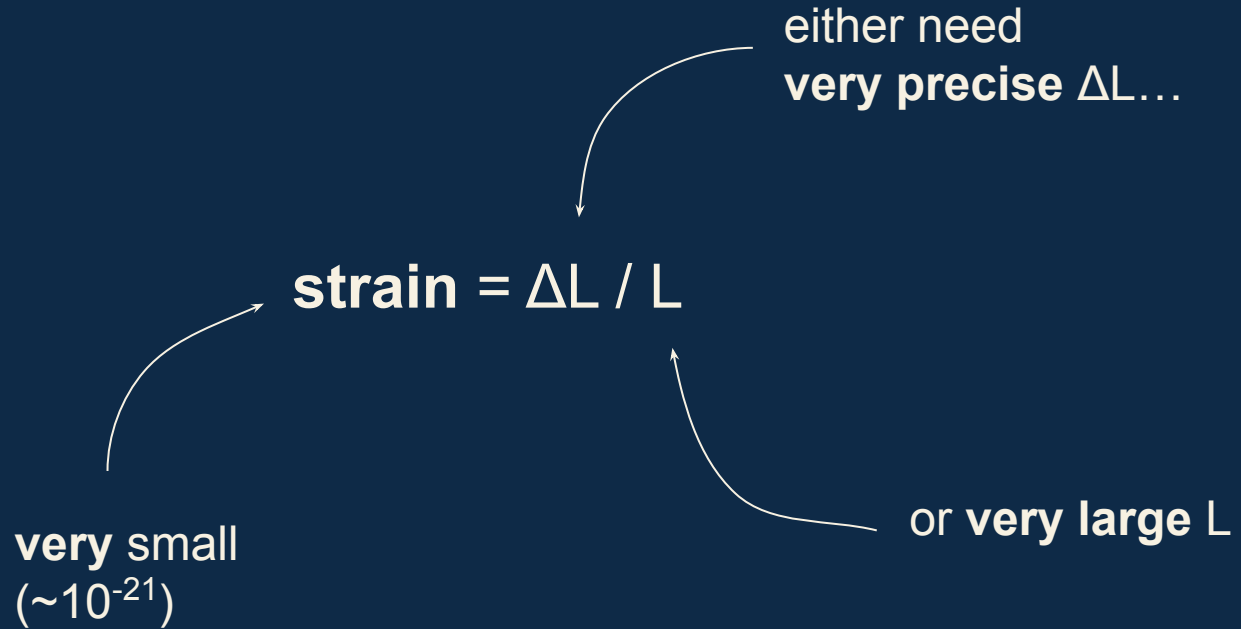
How do we detect GWs?



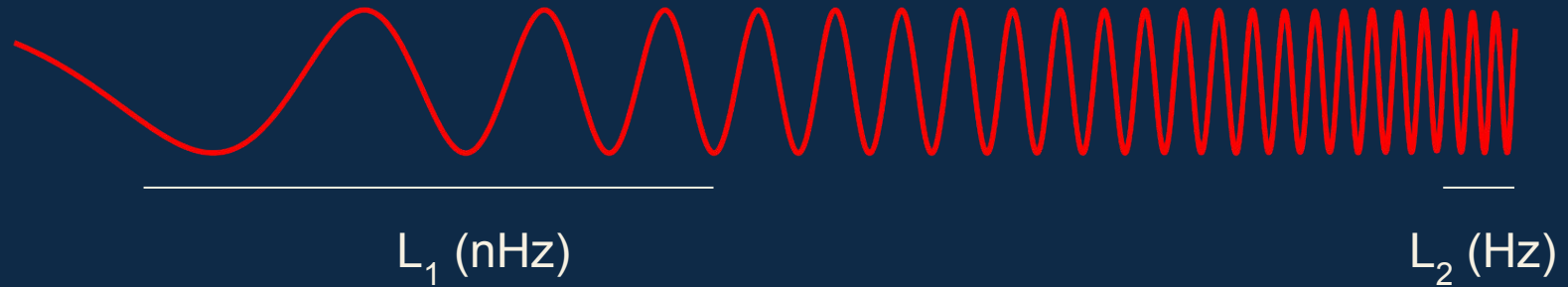
strain = $\Delta L / L$

very small
($\sim 10^{-21}$)

How do we detect GWs?



Observatory Size $\leftrightarrow$ Detectable Frequencies



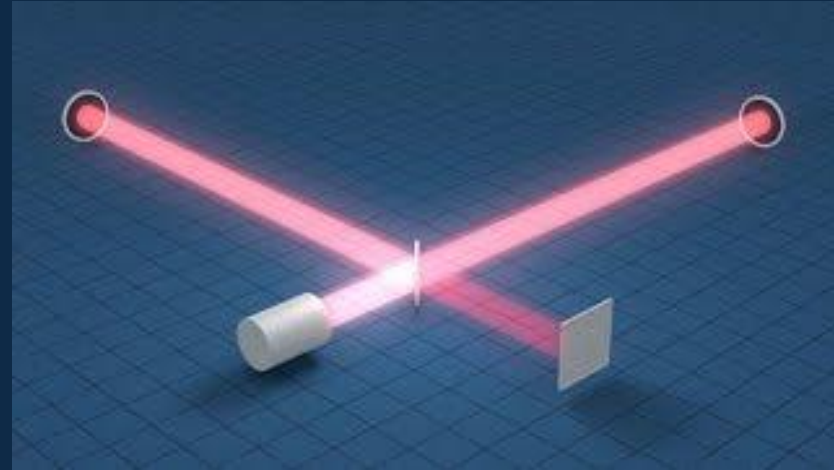
A Brief History of Gravitational Waves

1916 Einstein proposes GWs
as part of GR

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1960's Gertsenshtein & Pustovoit, Weiss et al.
independently propose **interferometric
detection**



A Brief History of Gravitational Waves

1916 Einstein proposes GWs
as part of GR

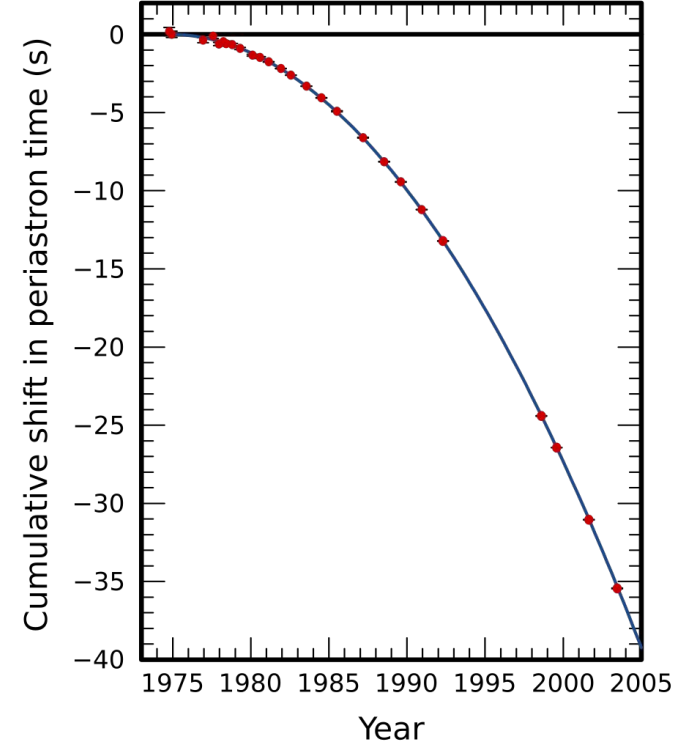
1960's Gertsenshtein & Pustovoit, Weiss et al.
independently propose **interferometric
detection**

1969 Joseph Weber claims GW detection
with **resonant bar detector**, but result
cannot be reproduced



A Brief History of Gravitational Waves

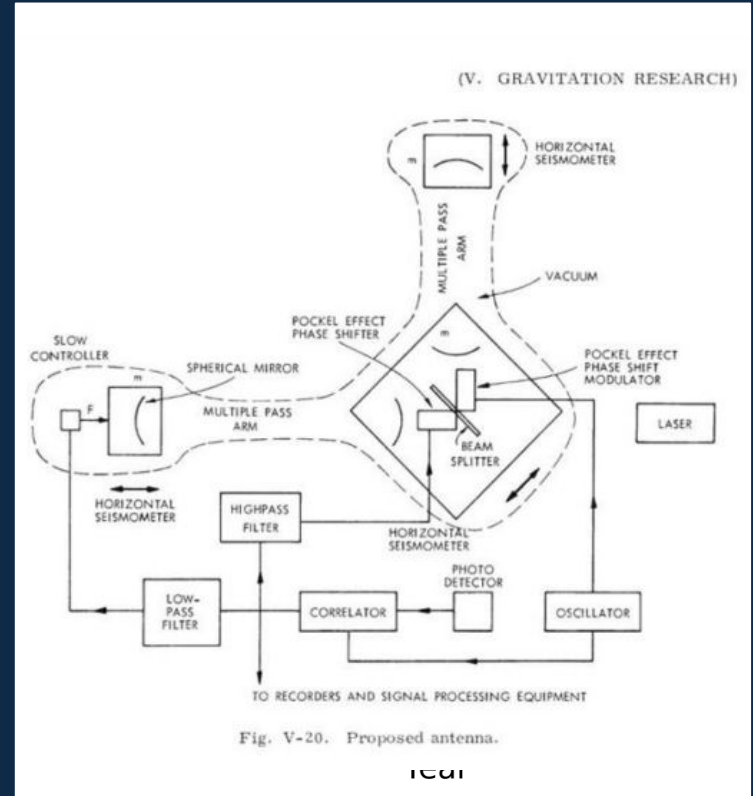
1978 **Hulse-Taylor Binary:** first
indirect detection of GWs



A Brief History of Gravitational Waves

1978 **Hulse-Taylor Binary:** first *indirect* detection of GWs

1984 **LIGO** founded

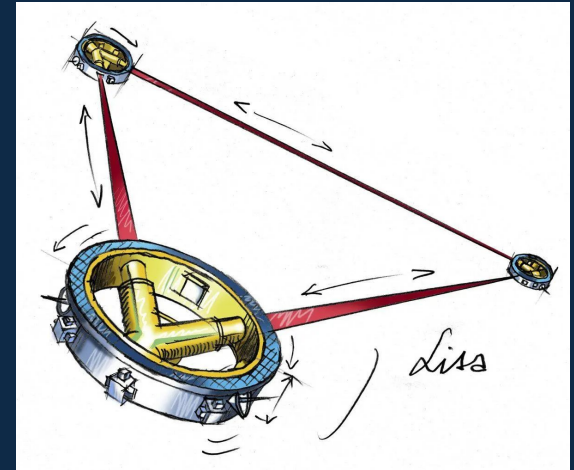


A Brief History of Gravitational Waves

1978 **Hulse-Taylor Binary:** first
indirect detection of GWs

1984 **LIGO** founded

1990s **LISA** proposed



A Brief History of Gravitational Waves

1978 **Hulse-Taylor Binary:** first *indirect* detection of GWs

1984 **LIGO** founded

1990s **LISA** proposed

2001 Beginning of initial LIGO operations



A Brief History of Gravitational Waves

2005 NANOGrav begins **pulsar timing array observations**

2014 **Another false alarm detection claim** via CMB measurements

A Brief History of Gravitational Waves

Caltech

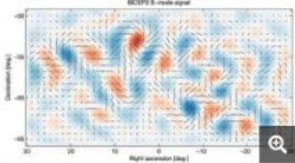
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BICEP2 Discovers First Direct Evidence of Inflation and Primordial Gravitational Waves

March 17, 2014

Astronomers announced today that they have acquired the first direct evidence that gravitational waves rippled through our infant universe during an explosive period of growth called inflation. This is the strongest confirmation yet of cosmic inflation theories, which say the universe expanded by 100 trillion trillion times in less than the blink of an eye.



Gravitational waves from inflation generate a faint but distinctive twisting pattern in the polarization of the CMB, known as a "curl" or B-mode pattern. Shown here is the actual B-mode pattern

BICEP2 (Background Imaging of Cosmic Extragalactic Polarization), March 2014

nature

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Published: 30 January 2015

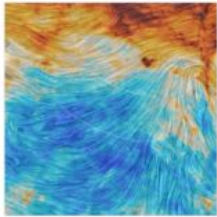
Gravitational waves discovery now officially dead

Ron Cowen

[Nature \(2015\)](#) | [Cite this article](#)

1546 Accesses | 4 Citations | 681 Altmetric | [Metrics](#)

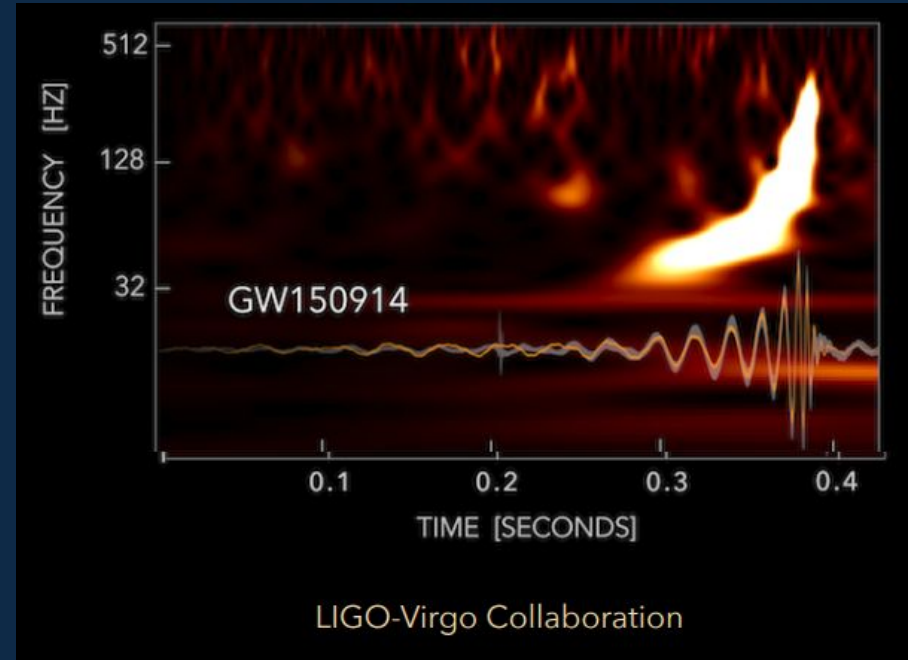
Combined data from South Pole experiment BICEP2 and Planck probe point to Galactic dust as confounding signal.



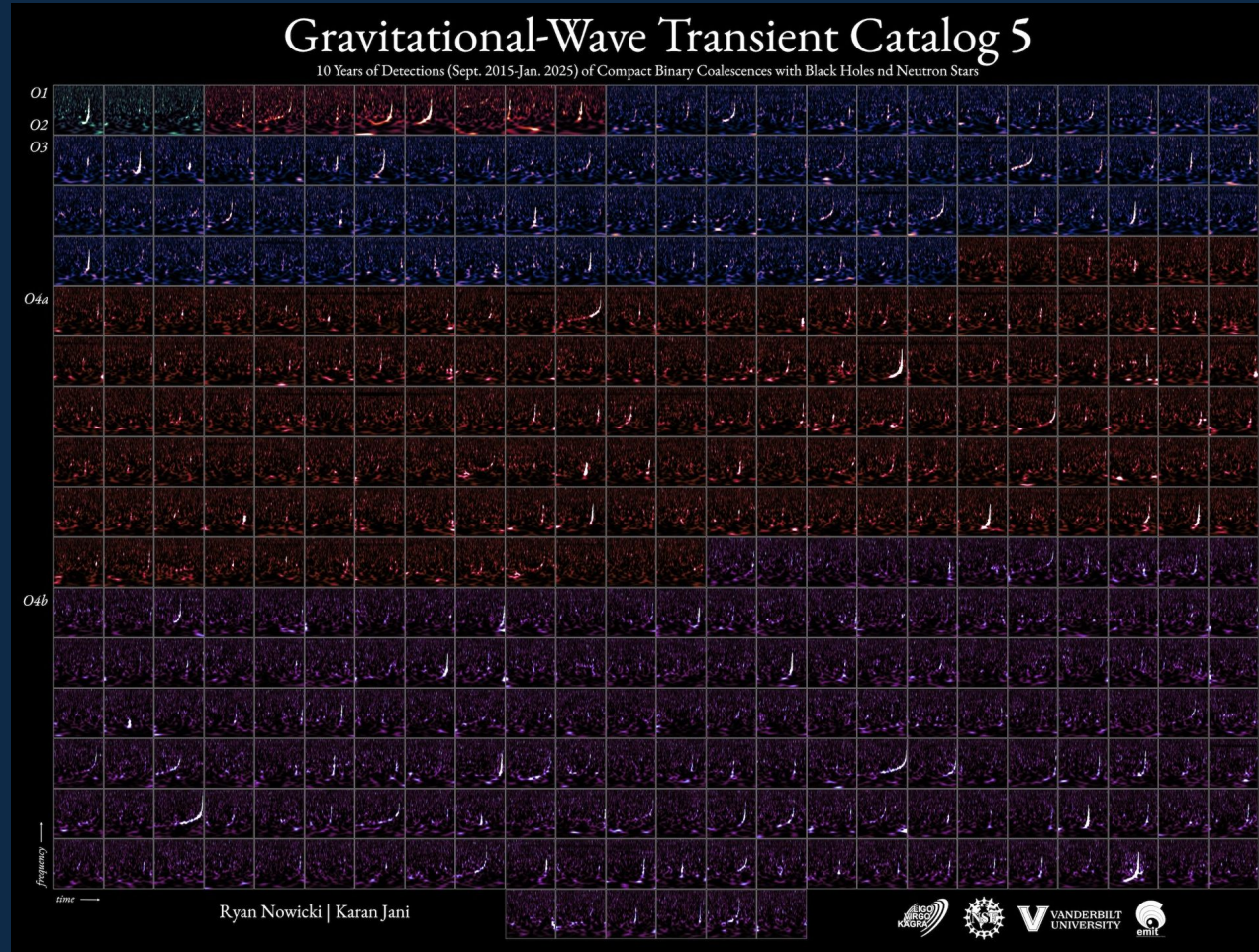
Nature, January 2015

A Brief History of Gravitational Waves

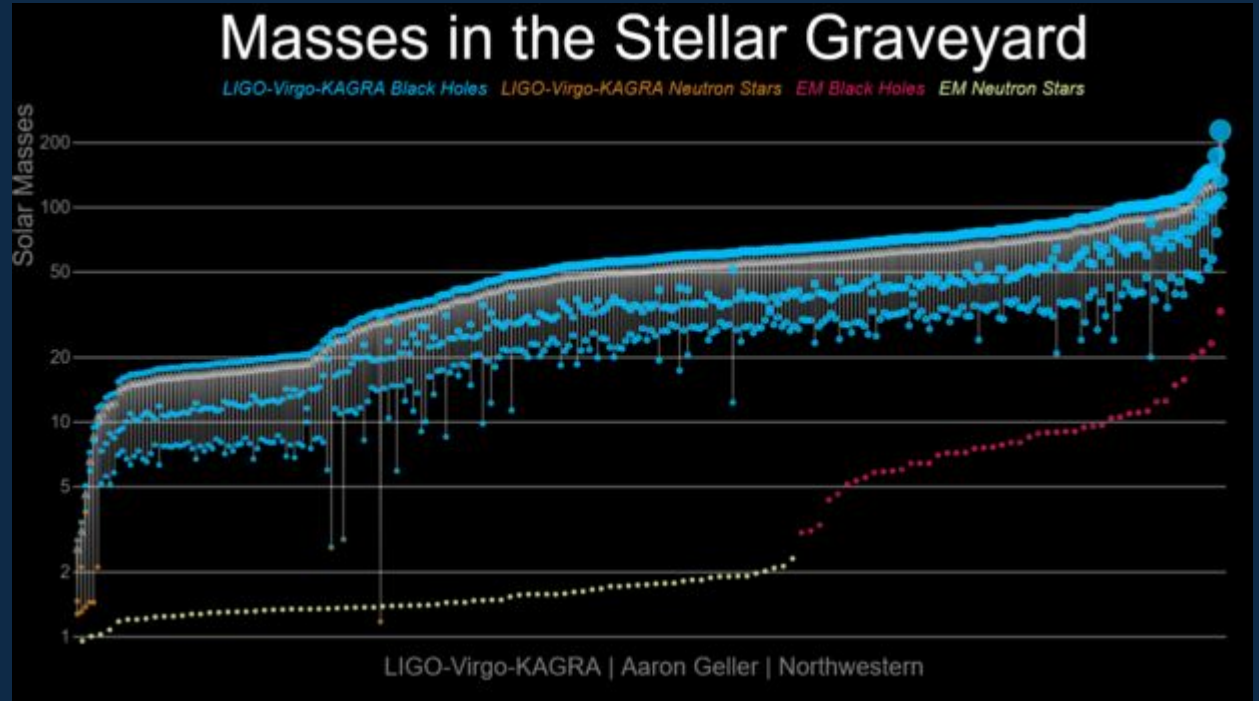
2015 First direct detection of GWs!



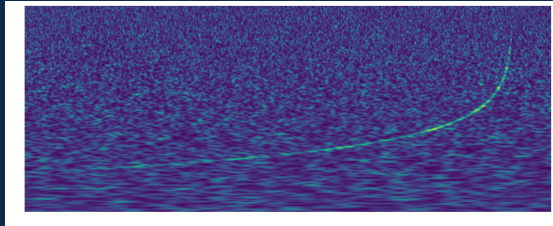
Ten Years Later



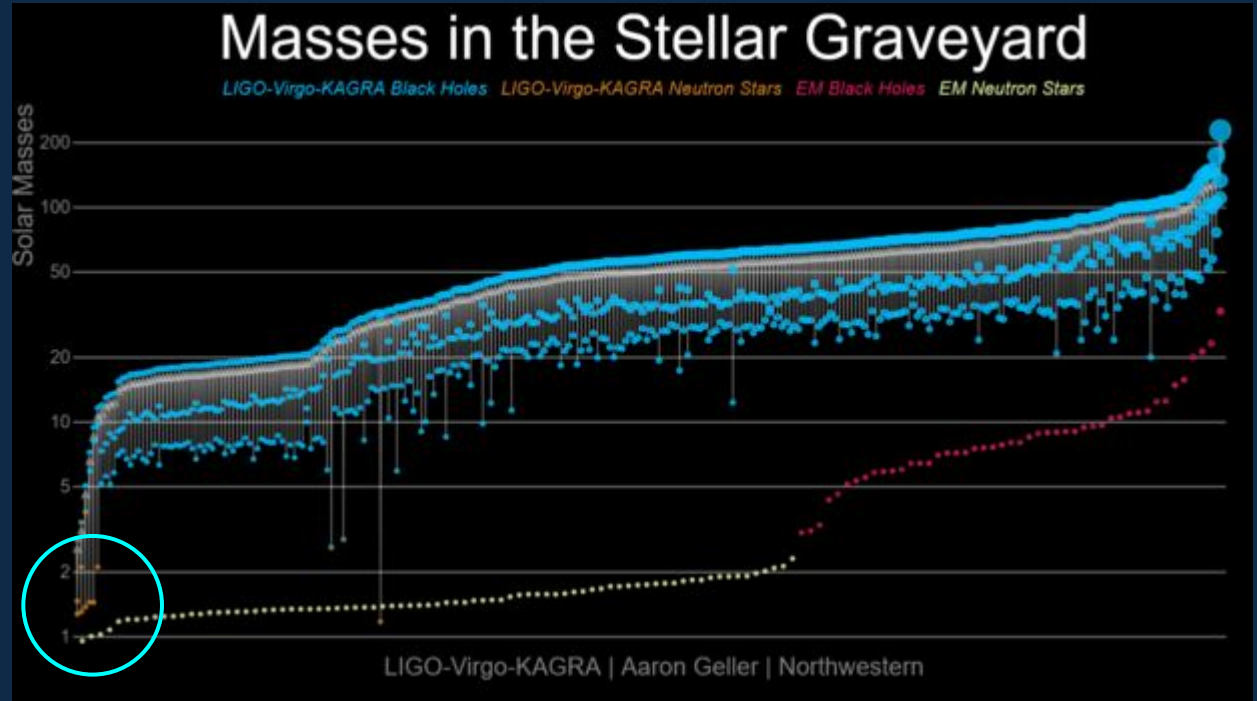
Ten Years Later



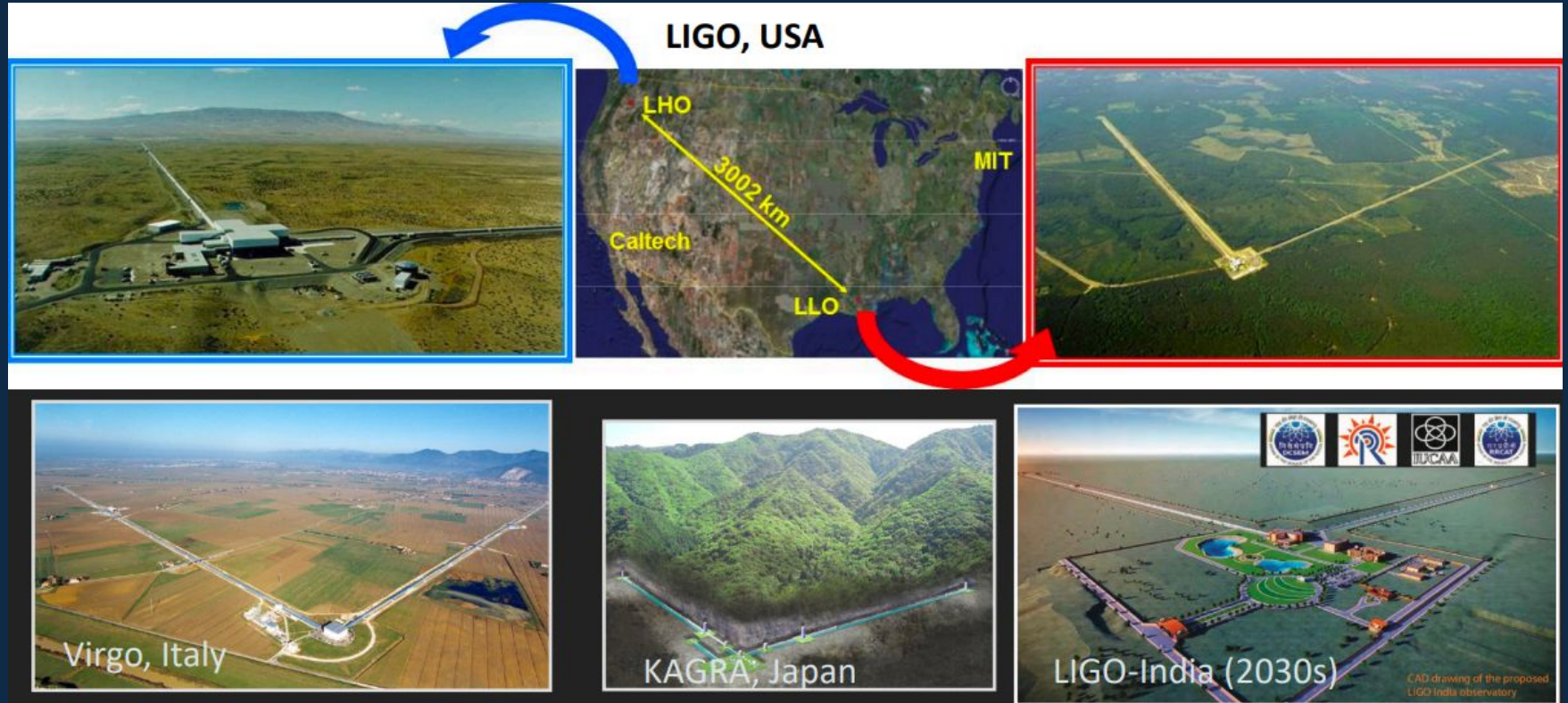
Ten Years Later



GW170817:
first **multimessenger**
GW observation



Current Terrestrial GW Detector Network



Taking a step back

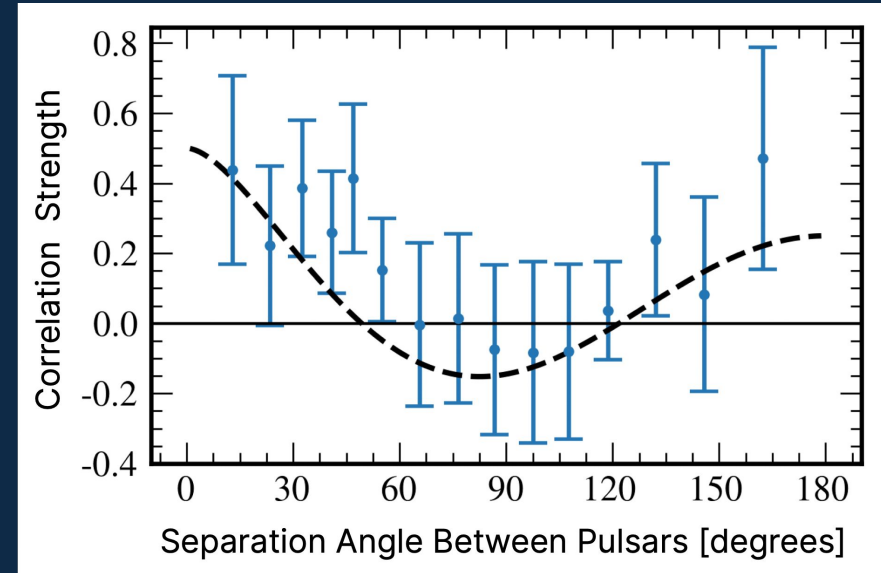
- 1970s** **Pulsar timing arrays**
proposed by Sazhin and Detweiler
- 1983** **Hellings & Downs** develop framework
for background correlations
- 2005** NANOGrav begins **pulsar
timing array observations**
- 2020** Hints of a common
process...



A Brief History of Gravitational Waves, continued

2023

**Detection of (strong evidence for)
a GW Background in PTAs**



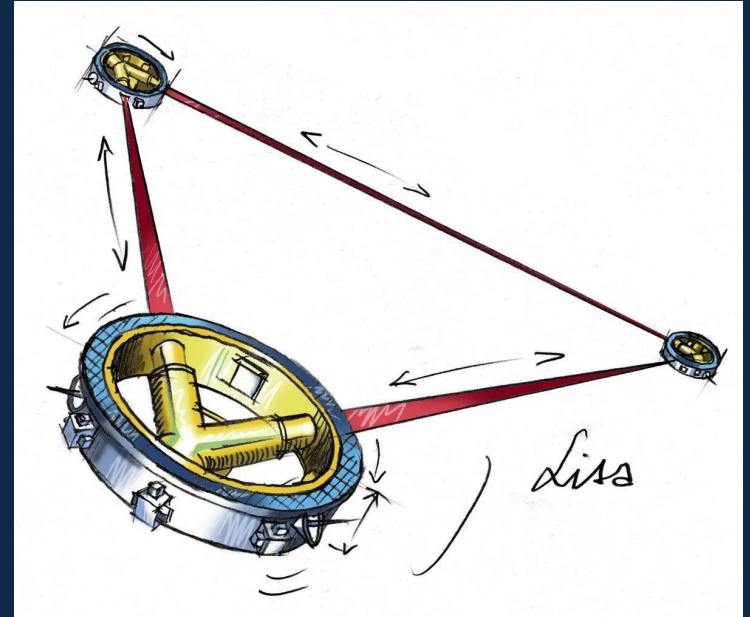
Meanwhile...

1990s The **Laser Interferometer Space Antenna (LISA)** proposed

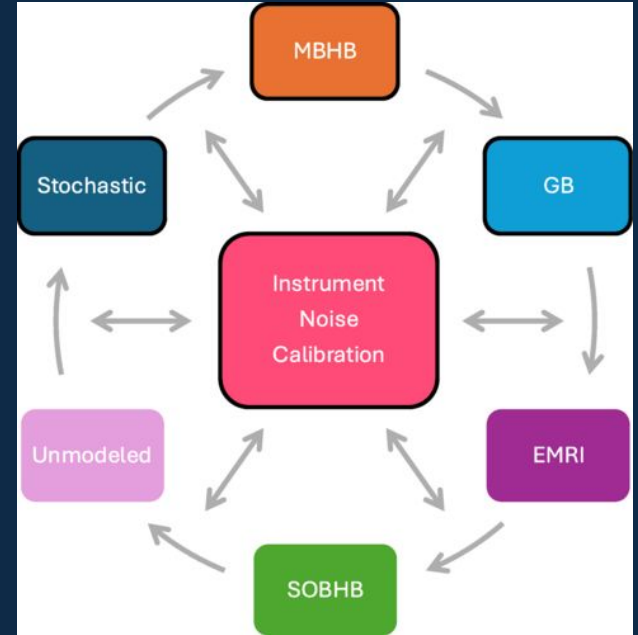
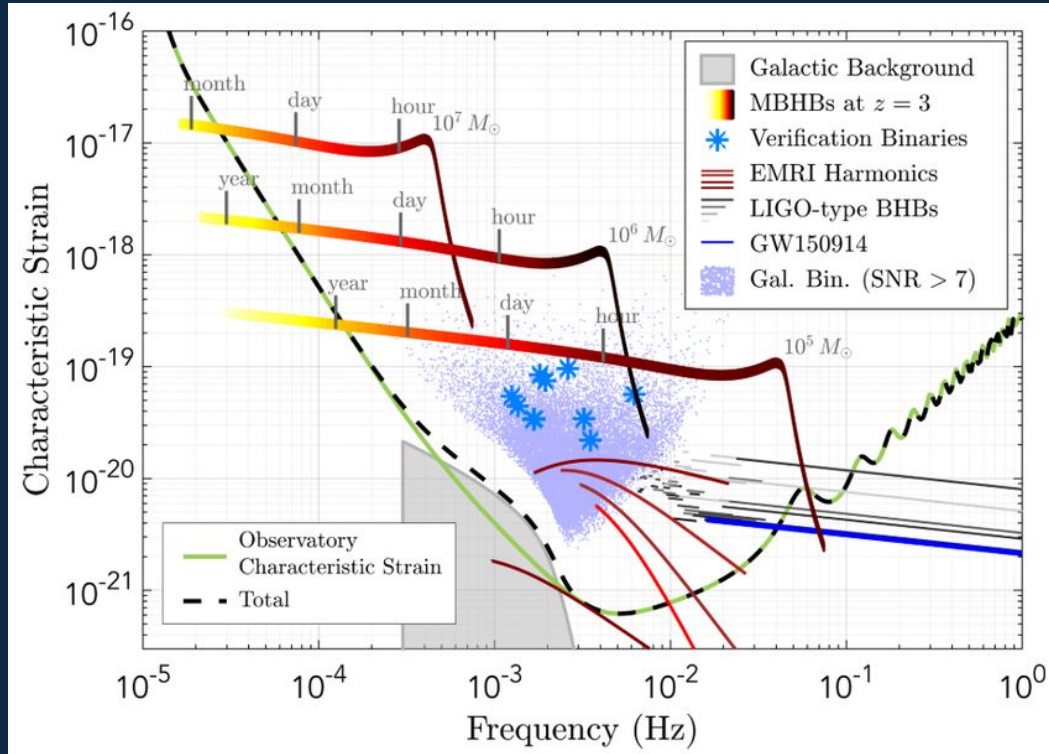
2000s+ Funding drama

2015 **LISA Pathfinder** launches, 2-year mission is a great success!

2025 LISA formally adopted as a joint ESA-NASA mission, with a launch date of **2035**



Biggest Challenge for LISA: *too many gravitational waves*

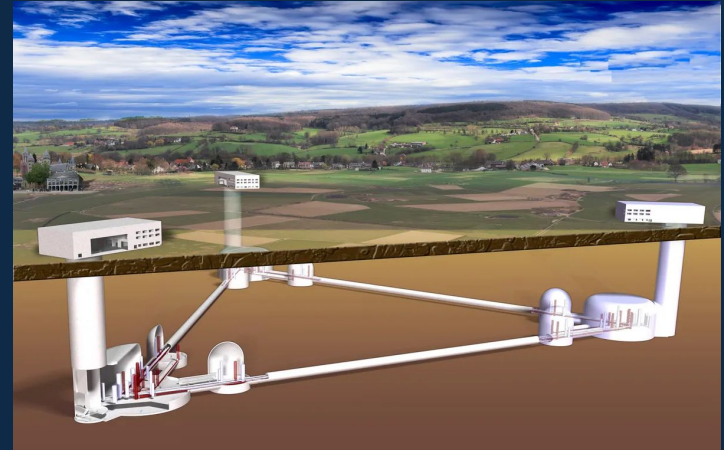


Katz+25

Looking to the future!



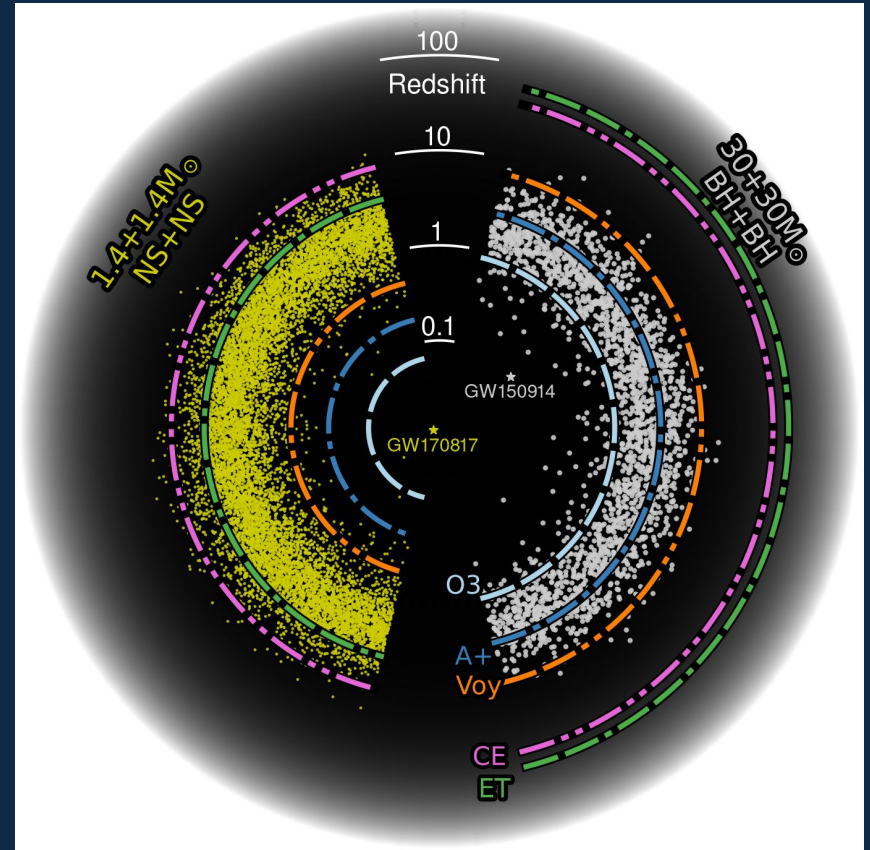
Cosmic Explorer (US)



Einstein Telescope (EU)

Looking to the future!

The 3G pitch:
every stellar-mass compact binary
merger **in the observable universe**



microHz and deciHz:

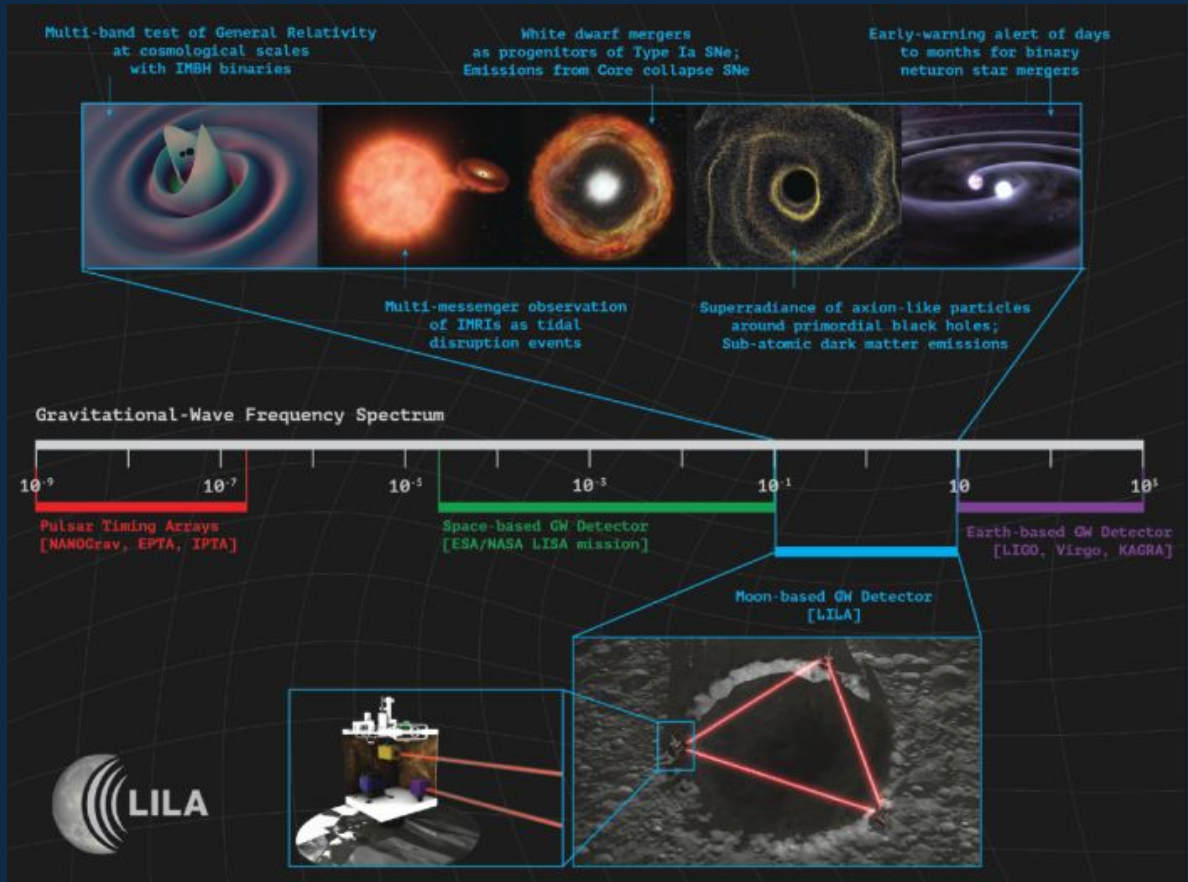
Proposed concepts to expand across the GW spectrum

deciHz

Laser
Interferometer
Lunar
Antenna

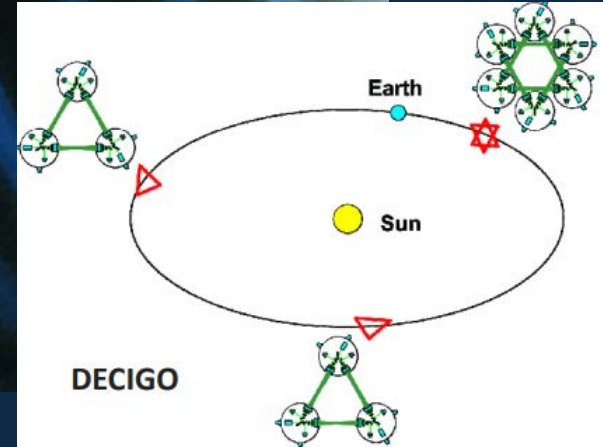
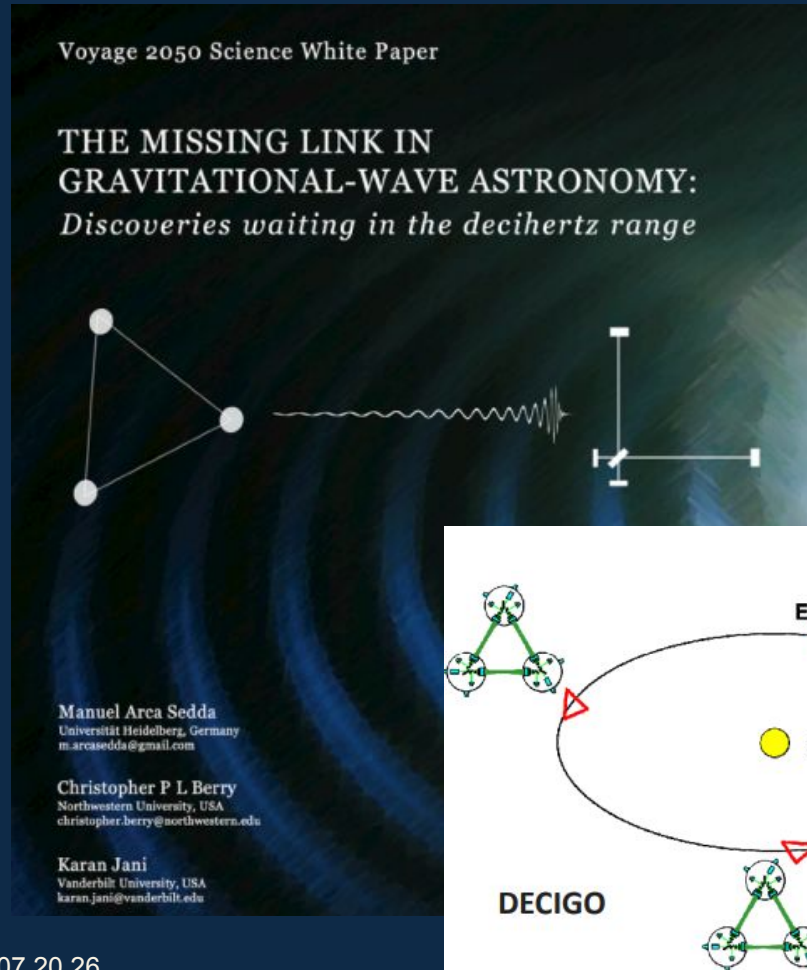
and

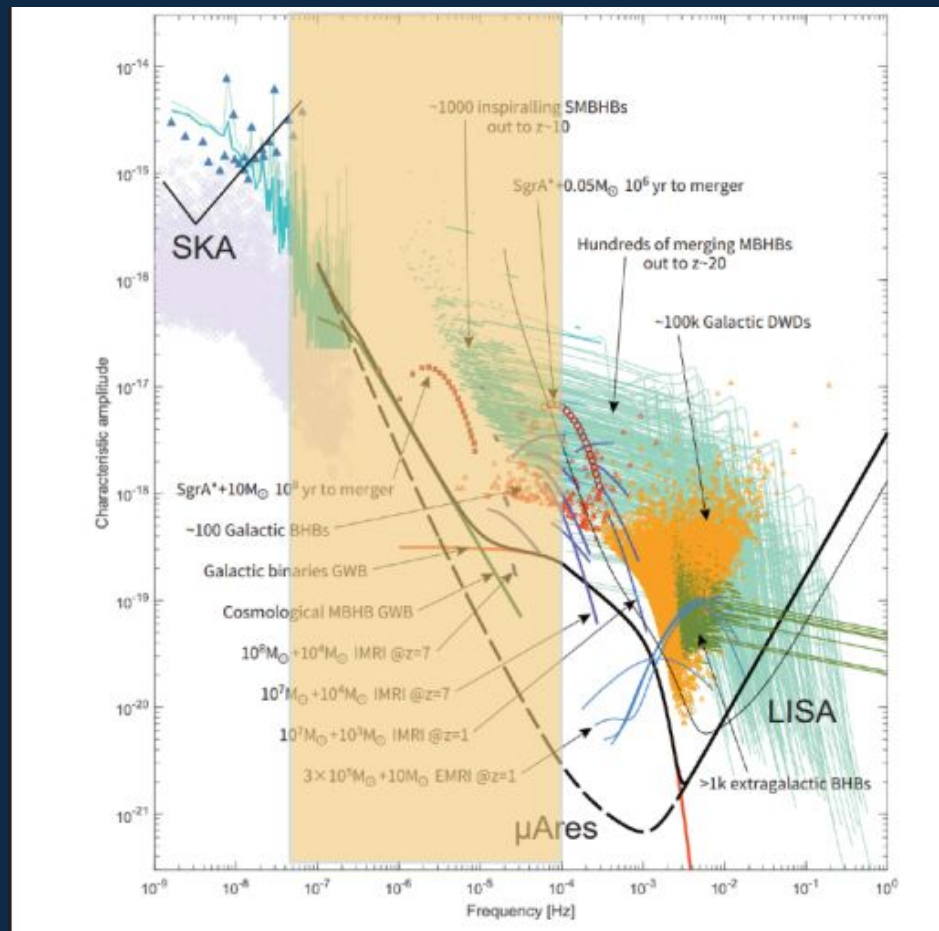
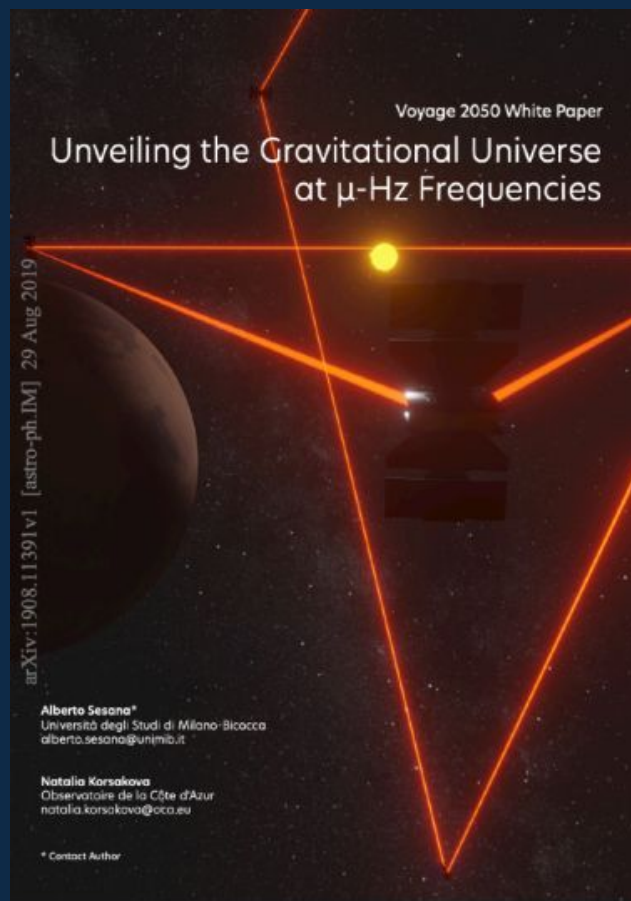
Lunar
Gravitational
Wave
Antenna

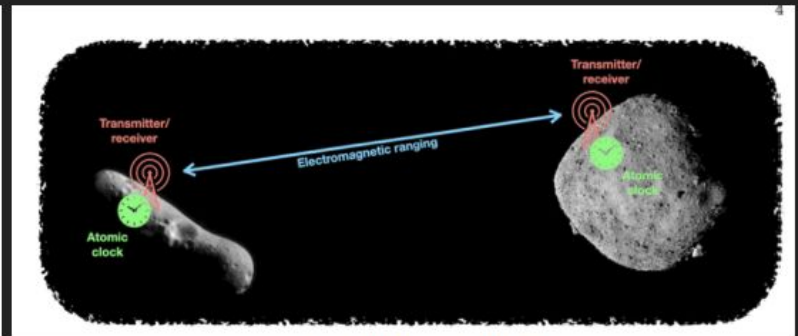
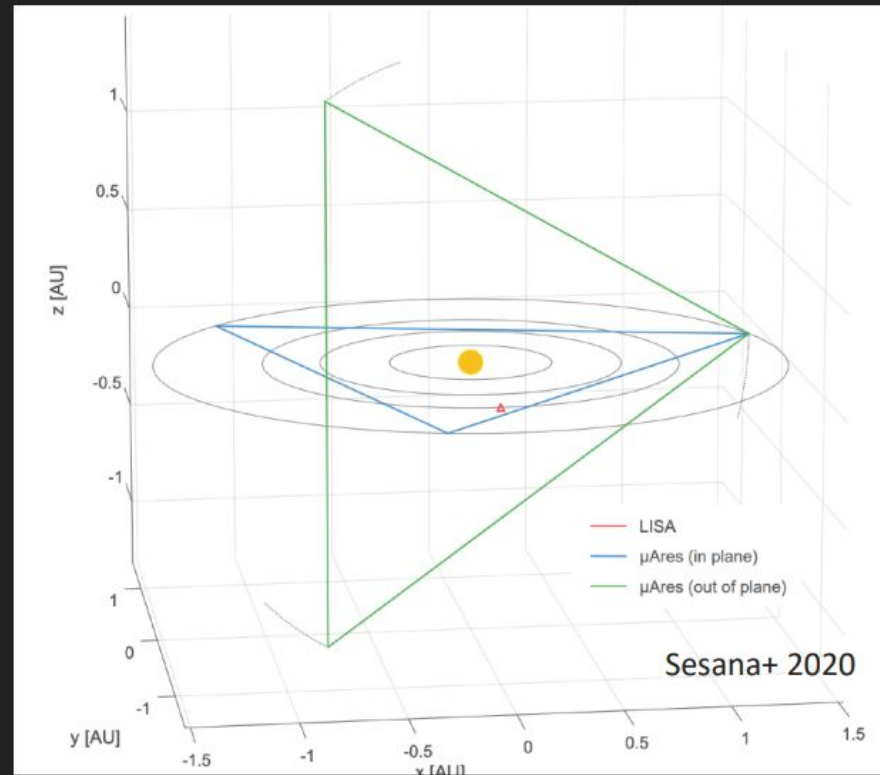


deciHz

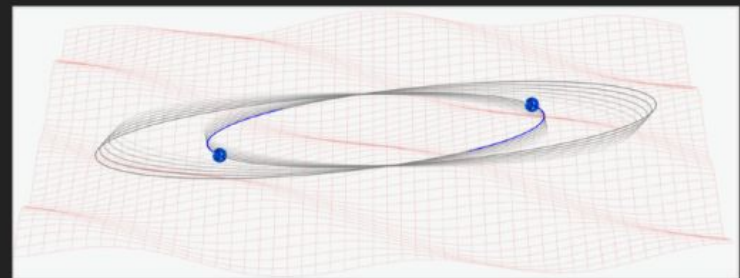
Deci-Hz
Interferometer
Gravitational
Wave
Observatory
(DECIGO)







Fedderke+ 2022



Blass & Jenkins 2022

Summary

- Just as we did with electromagnetic observations, we are expanding our ability to do science **across the GW spectrum!**
- **Right now:** LIGO/Virgo/KAGRA are observing hundreds of stellar-mass compact binary mergers. Pulsar timing arrays are observing a nHz GW background – and hunting for individual supermassive black hole binaries!
- **Near future:** LISA will observe GWs in the mHz, opening a new observational window into merging massive black hole binaries, Galactic white dwarf binaries, extreme mass ratio inspirals and more.
- **Far future:** Cosmic Explorer and Einstein Telescope observe every compact binary merger in the observable universe. New concepts give rise to new observations across the GW spectrum!

Questions?

